Cloud Computing

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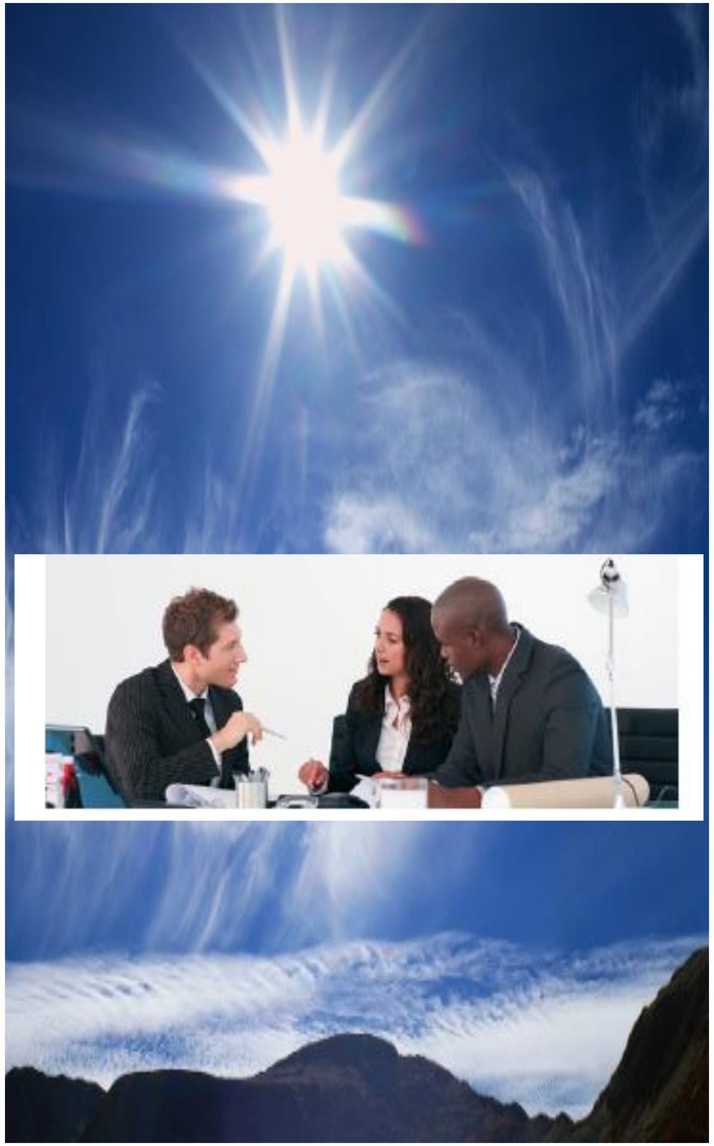


Objectives

After completing this unit, you should be familiar with:

- Cloud computing definition
- Describing cloud computing in one sentence
- Factors that lead to the adoption of cloud computing
- Explaining cloud concepts such as, infrastructure as a service, platform as a service, and software as a service
- Business benefits of cloud computing for IT, application development, and testing
- Describing cloud computing deployment models
- Identifying cloud computing adoption risks
- Differentiating between traditional IT and cloud computing services.

Introduction to Cloud Computing

- Cloud Computing Market Report Scope:

Attribute	Details
Market size value in 2024	USD 752.44 billion
Revenue forecast in 2030	USD 2,390.18 billion
Growth Rate	21.2% from 2024 to 2030
Forecast period	2024 - 2030
Report updated	May 2024
Quantitative units	Revenue in USD million/billion from 2024 to 2030
Report coverage	Revenue forecast, company share, competitive landscape, growth factors, trends
Regional scope	North America; Europe; Asia Pacific; Latin America; MEA
Key companies profiled	Alibaba Cloud; Amazon Web Services, Inc. ; Google; IBM; Microsoft Azure; Oracle Cloud; Inc.; Salesforce, Inc.; and VMware LLC

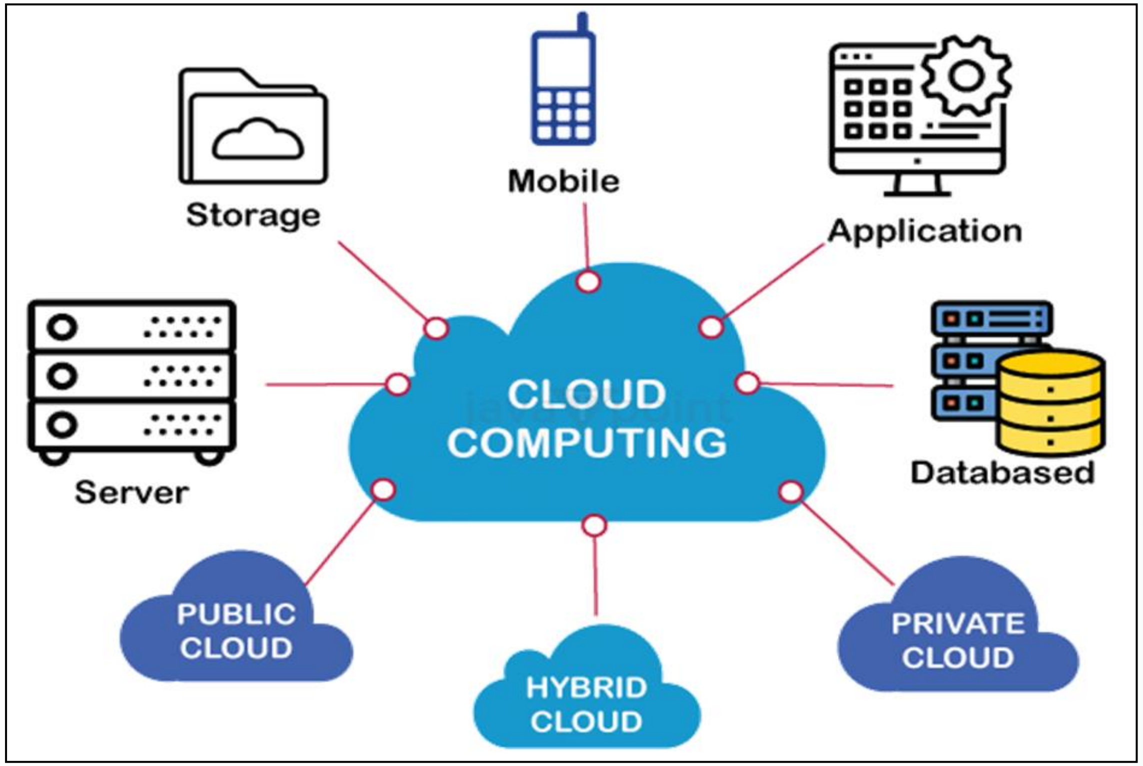
- 80% of the US economy is driven by service industry, leaving only 15% by manufacturing and 5% from the agriculture.

Introduction to Cloud Computing

Why Cloud Computing?:

- Companies, usually follows the **traditional methods to provide the IT infrastructure.** That means company need a server room which is the basic need of IT.

- In server room, there should be a **database server, mail server, networking, firewalls, routers, modem, switches, high net speed, and the maintenance engineers.**



- To establish such IT infrastructure, we need to spend lots of **money.** To overcome all problems and to reduce the IT infrastructure cost.

Introduction to Cloud Computing

Why Cloud Computing?:

- **IT provisioning: Real world estimates**

  - Average server utilization is 5% to 20%.
  - Peak workload exceeds the average by factors of 2 to 10.
  - Peak loads may occur based on the time of day or based on other factors (e.g. photo sharing after the holidays, drop/add within two weeks of start of term, etc.)

Introduction to Cloud Computing

Cloud Computing Definitions:

- **Cloud Computing:** Is the delivery of hosting services that are provided to a client over the Internet. In simple term, it means storing, managing, and accessing the data & programs on the remote servers that are hosted on internet instead of computer's hard drive

- **Cloud Computing:** Is the on-demand availability of computer resources (data storage - computing power) without active management of the users. In short, we store, manage, and process data on remote servers.

- **Cloud Computing:** "Is a model for enabling convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction". **(NIST Definition)**

Introduction to Cloud Computing

Cloud Computing Definitions:

- The **key points** in this **definition** include the following:
  - **Cloud computing is not so much a technology as a model for delivering services.**
  - Cloud computing enables users to **easily access IT services** like networks, servers, storage, applications, and services.
  - **Network connectivity is a requirement** for easy, on-demand access to cloud resources.
  - **Rapid resource provisioning** and reclamation fall under the rapid **elasticity** characteristic of cloud computing, but on- demand self-service is distinguished by low management effort and service provider interaction.

Cloud Computing Characteristics

- **Agility:** The **cloud** works in a **distributed computing environment**. It shares resources among users and works very fast.
- **High availability and reliability:** The availability of servers is high and more reliable because the chances of infrastructure failure are minimum.
- **High Scalability:** Cloud offers "on-demand" provisioning of resources on a large scale, without having engineers for peak loads.
- **Multi-Sharing:** With the help of cloud computing, multiple users and applications can work more efficiently with cost reductions by sharing common infrastructure.
- **Maintenance:** Maintenance of cloud computing applications is easier, since they do not need to be installed on each user's computer and can be accessed from different places. So, it reduces the cost also.

Cloud Computing Characteristics

- **Device and Location Independence:** CC enables the users to access systems using a web browser regardless of their location or what device they use e.g. PC, mobile phone, etc. As infrastructure is off-site (typically provided by a third-party) and accessed via the Internet, users can connect from anywhere.

- **Low Cost:** by using cloud computing, the cost will be reduced because to take the services of cloud computing, IT company need not to set its own infrastructure and pay-as-per usage of resources.

- **Services in the pay-per-use mode:** Application Programming Interfaces (APIs) are provided to the users so that they can access services on the cloud by using these APIs and pay the charges as per the usage of services.

Features of Cloud Computing

- **It is elastic:** Cloud computing is flexible in nature, where users can **scale up** and **scale down** the resources as needed.
- **Pay per use:** Usage is metered and user **pays only for how much the resources were used**.
- **Operation:** The services are completely **handled by the provider**.
- **Reduce capital cost:** No need to **invest money** on purchasing and maintaining of hardware and software, software licensing, training required for IT staff.
- **Remote accessibility:** Users can access **applications and data stored on cloud** from anywhere any time worldwide through a device with internet connection.
- **Better use of IT staff:** Staff with in enterprise need not worry on purchasing and maintaining of servers, software, up gradation of servers and software, software licensing etc., instead they can concentrate more on work.

Features of Cloud Computing

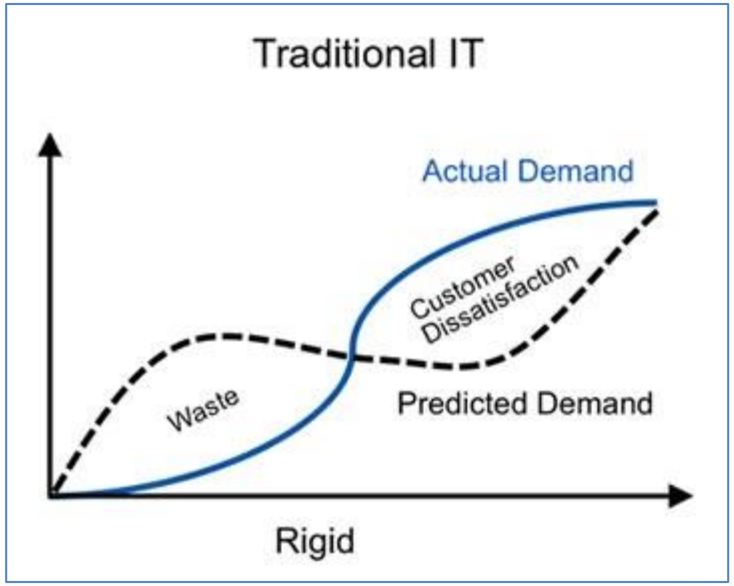
Elasticity in Cloud Computing

- ❑ In the traditional IT what happened we used to provision or dimension our resources for some capacity which would be stagnant over a period of time.

- ❑ To increase this capacity we have to scale up our systems accordingly so usually we have to over dimension our resources.

- ❑ Elasticity means "how rapidly we can configure the capacity to meet the demand (**all about demand and capacity**)".

- ❑ In cloud computing, elasticity is defined as "the degree to which a system is able to adapt to workload changes by provisioning and de-provisioning resources in an autonomic manner, such that at each point in time the available resources match the current demand as closely as possible"



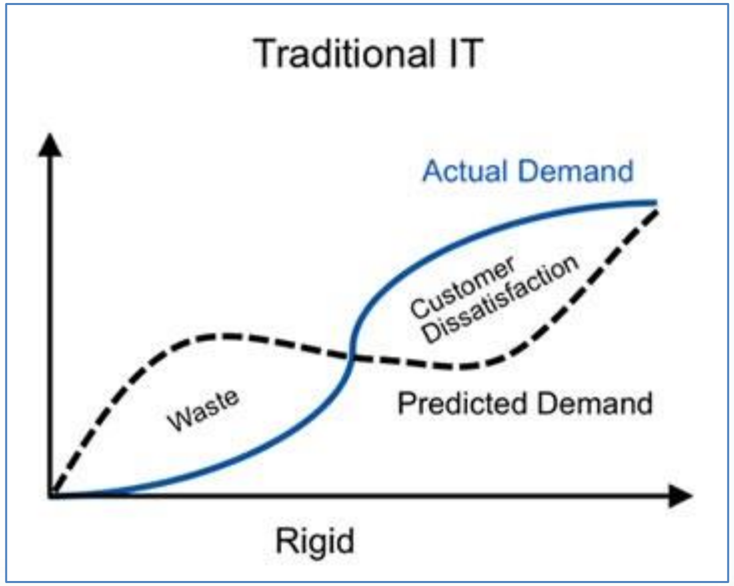
Features of Cloud Computing

Elasticity in Cloud Computing

- ❑ Elasticity is a defining characteristic that differentiates cloud computing from previously proposed computing paradigms, such as grid computing.

- ❑ The dynamic adaptation of capacity, e.g., by altering the use of computing resources, to meet a varying workload is called "elastic computing".

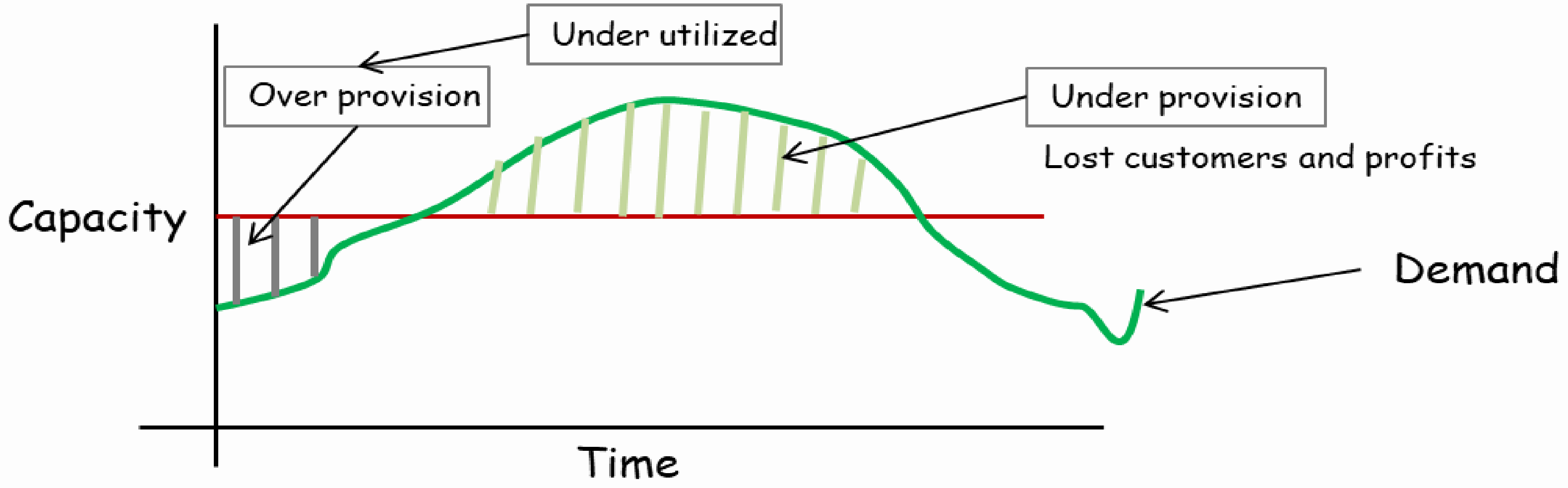
- ❑ However the demand is not fixed so suppose let's take a case of web services provider usually what will happen in low traffic period and it will gradually increase and we would have some spikes there and then it will decrease to a normal period.



Features of Cloud Computing

Elasticity in Cloud Computing

- We have two factors (**Capacity and Demand**):
  - **Over provisioning:** the capacity is under utilized and the area is the area of under provisioning
  - **Under Provisioning:** capacity is fully utilized but we have loss of customer and traffic.
  - Now its required to configure our resources to match this pattern of the demand.
  - So suppose the demand is increasing up in a upward direction we could have provision the resources accordingly and when the demand is increasing
  - We could have decreased the capacity or the resources
  - This factor is called elasticity



Features of Cloud Computing

❑ IT provisioning: (Over-provisioning)

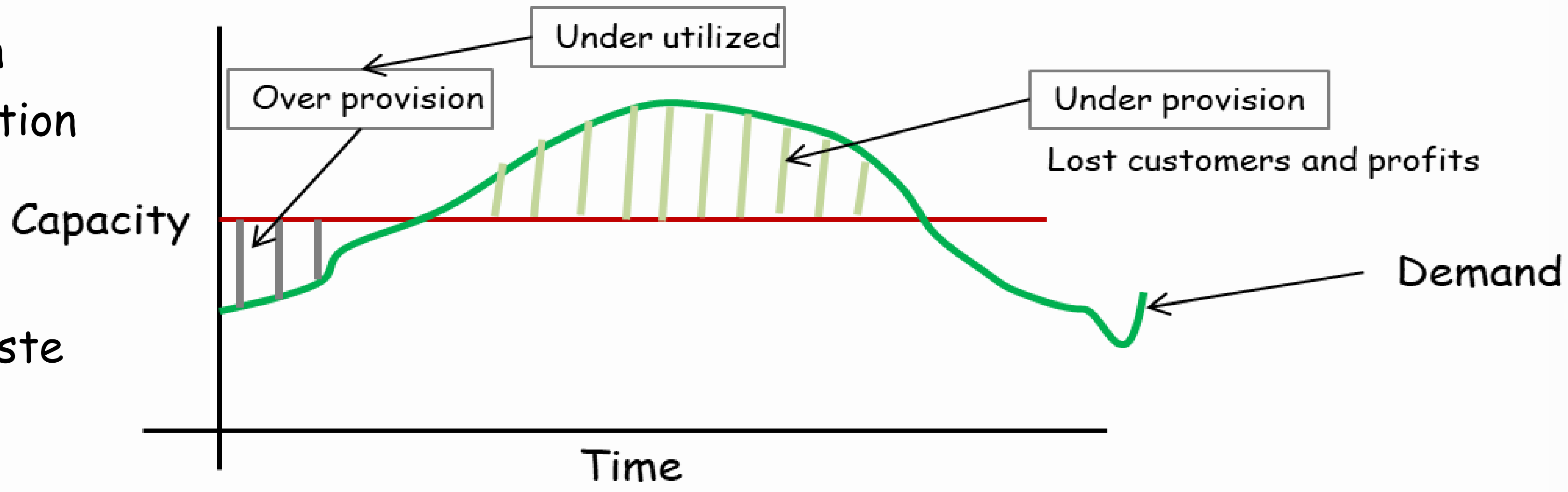
❖ Refers to the practice of allocating more computing resources (such as CPU, memory, and storage) to a system or network than what is currently needed by its workload. This strategy aims to ensure that sufficient resources are available to handle unexpected spikes in demand without compromising performance.

❖ **Benefits:**

- Increased performance
- Enhanced data protection
- Improved capacity utilization

❖ **Drawbacks:**

- Higher initial cost
- Potential for resource waste



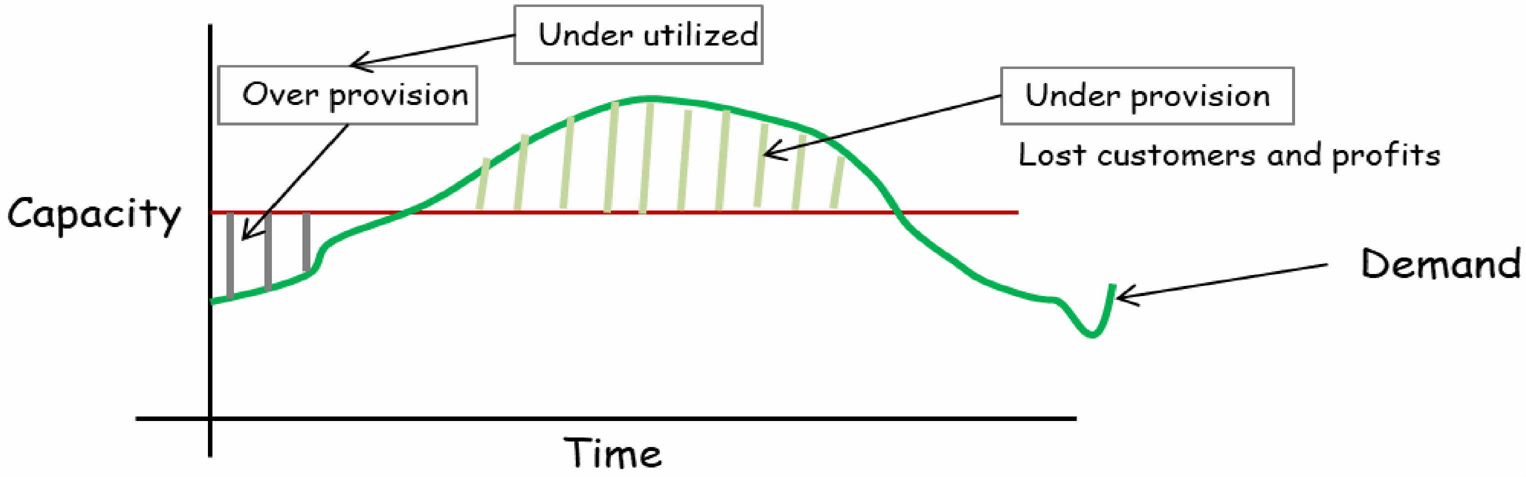
Features of Cloud Computing

❑ IT provisioning: (Under-provisioning)

❖ Refers to the practice of allocating fewer computing resources than necessary to an application or system. This can lead to poor performance, slow response times, and reduced availability, ultimately affecting user experience and the overall effectiveness of your digital infrastructure.

❖ **Drawbacks:**

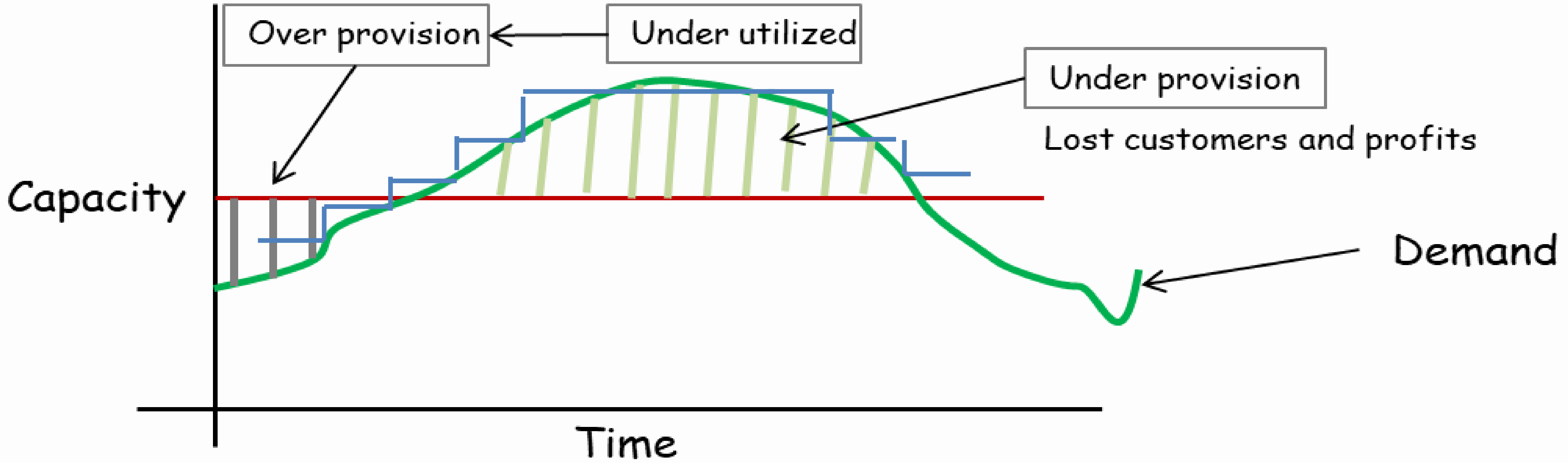
- ➢ Performance degradation
- ➢ Unreliable availability
- ➢ Reduced productivity
- ➢ Increased costs



Features of Cloud Computing

❑ IT provisioning: (Dynamic-provisioning)

❖ Refers to a strategy that efficiently manages server resources by activating only the necessary number of resources to meet the current and future demand, while putting unused servers in sleep mode.



Cloud Computing Advantages

- **Cost Savings:** Pay for what you use, with no upfront infrastructure costs.
- **Scalability:** Easily scale resources up or down based on demand.
- **Flexibility:** Access resources and applications from anywhere with an internet connection.
- **Enhanced Data Security:** CC guarantee that data is handled and stored safely, cloud service providers offer cutting-edge security features like encryption, access limits, and regular security audits. Businesses can rest easy knowing that their important data is secure.
- **Data Backup and Restoration:** CC offers a quick and easy method for data backup and restoration. Businesses may simply access and restore their data in the event of any data loss or system failure by keeping it in the cloud.

Cloud Computing Advantages

- **Improved Collaboration:** Collaboration is improved because cloud technologies make it possible for teams to share information easily. Multiple users may work together on documents, projects, and data thanks to shared storage in the cloud, enhancing productivity and teamwork.

- **Excellent Accessibility:** Users can access their data from anywhere in the world with an internet connection, making remote work, flexibility, and effective operations possible.

- **Cost-effective Maintenance:** Organizations using CC can save money on both H/w & S/W. Because cloud service providers manage the maintenance and updates, businesses no longer need to make costly infrastructure investments or set aside resources for continuous maintenance.

- **Reliability:** Cloud providers typically offer high uptime and data redundancy.

Cloud Computing Advantages

- **Mobility:** Cloud computing makes it simple for mobile devices to access data. Utilizing smartphones and tablets, users can easily access and control their cloud-based applications and data, increasing their mobility and productivity.

- **Pay-per-use Model:** CC uses a pay-per-use business model that enables companies to only pay for the services they really utilize. This method is affordable, eliminates the need for up-front investments, and offers budget management flexibility for IT.

- **Collaboration:** Enable seamless collaboration and data sharing among teams.

- **Scalable Storage Capacity:** Users can access their data from anywhere in the world with an internet connection, making remote work, flexibility, and effective operations possible.

Cloud Computing Advantages

- **Disaster Recovery and Business Continuity:** Cloud computing provides reliable options for these two issues. Businesses can quickly bounce back from any unforeseen disasters or disruptions thanks to data redundancy, backup systems, and geographically dispersed data centers.

- **Agility and Innovation:** Businesses can continue to be innovative and nimble thanks to cloud computing. Organizations may quickly embrace new solutions, test out emerging trends, and promote corporate growth with access to a variety of cloud-based tools, services, and technology.

- **Green Computing:** By utilizing technologies like virtualization and load balancing to maximize the use of computer resources, cloud providers can operate large-scale data centers built for energy efficiency, resulting in lower energy usage and a smaller carbon footprint.

Cloud Computing Disadvantages

- **Vendor Reliability and Downtime:** Because of technological difficulties, maintenance needs, or even cyber attacks, providers can face outages or downtime. Users may not be able to access their data or applications during these times, which affects business operations and productivity.

- **Limited Control and Customization:** Using standardized services and platforms offered by the cloud service provider is a common part of cloud computing. As a result, organizations may have less ability to customize and control their infrastructure, applications, and security measures. It may be difficult for some organizations to modify cloud services to precisely match their needs if they have special requirements or compliance requirements.

- **Data Security and Concerns about Privacy:** Data security and privacy concerns arise when sensitive data is stored on the cloud. Businesses must have faith in the cloud service provider's security procedures, data encryption, access controls, and regulatory compliance. Unauthorized access to data or data breaches can have serious repercussions, including financial loss, reputational harm, and legal obligations.

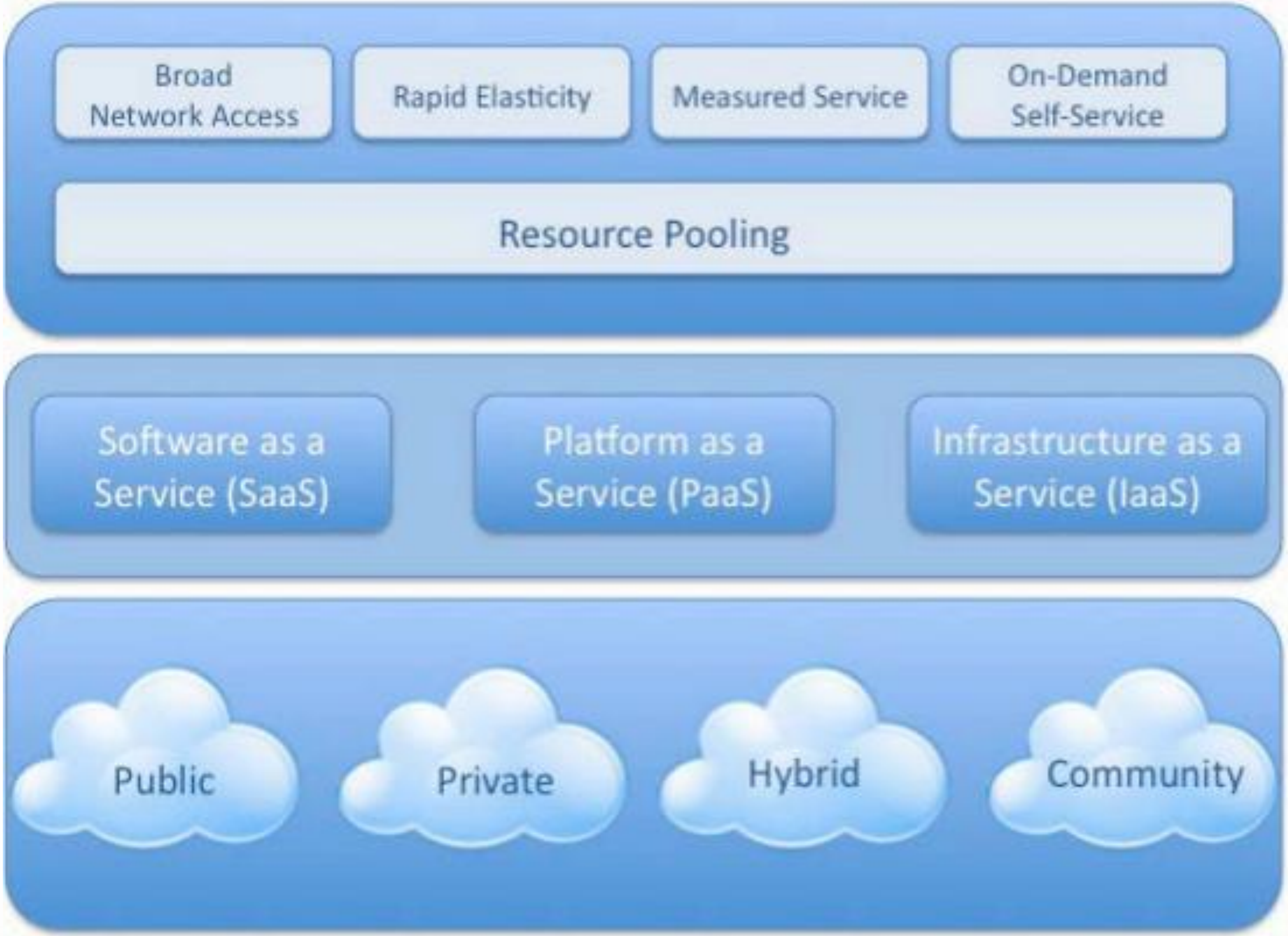
Cloud Computing Disadvantages

- **Hidden Costs and Pricing Models:** Although pay-as-you-go models and lower upfront costs make cloud computing more affordable, businesses should be wary of hidden charges. Data transfer fees, additional storage costs, fees for specialized support or technical assistance, and expenses related to regulatory compliance are a few examples.

- **Dependency on Service Provider:** When an organization depends on a cloud service provider, it is dependent on that provider's dependability, financial security, and longevity. Users may have disruptions and difficulties switching to alternate options if the provider runs into financial difficulties, changes their pricing policy, or even closes down their services.

- **Data Location and Compliance:** When data is stored in the cloud, it frequently sits in numerous data centers around the globe that may be governed by multiple legal systems and data protection laws. This may pose compliance issues, especially if some sectors of the economy or nations have stringent data sovereignty laws.

Visual Model of Cloud Computing



Computing power

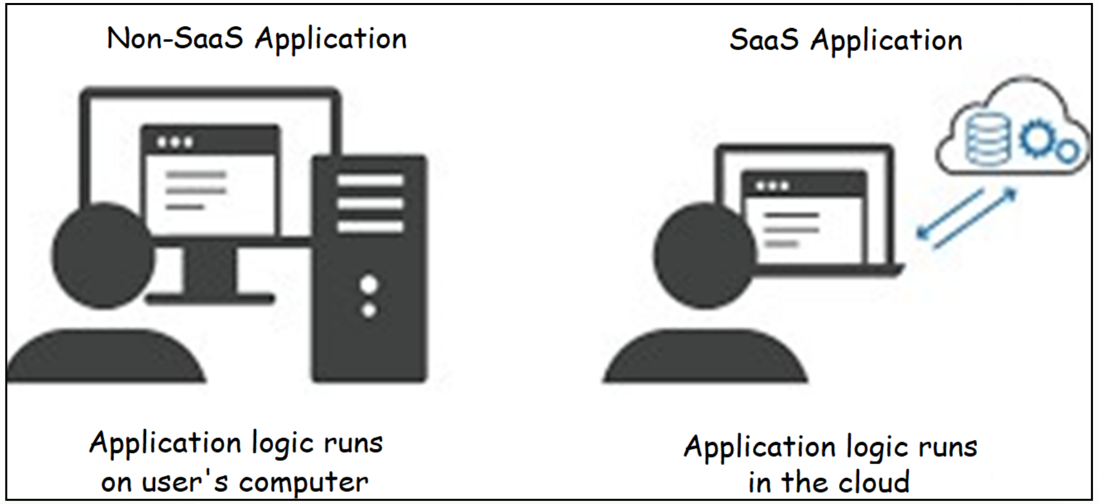
Service Models

Deployment Models

Cloud Service Models

❑ Software-as-a-Service (**SaaS**)

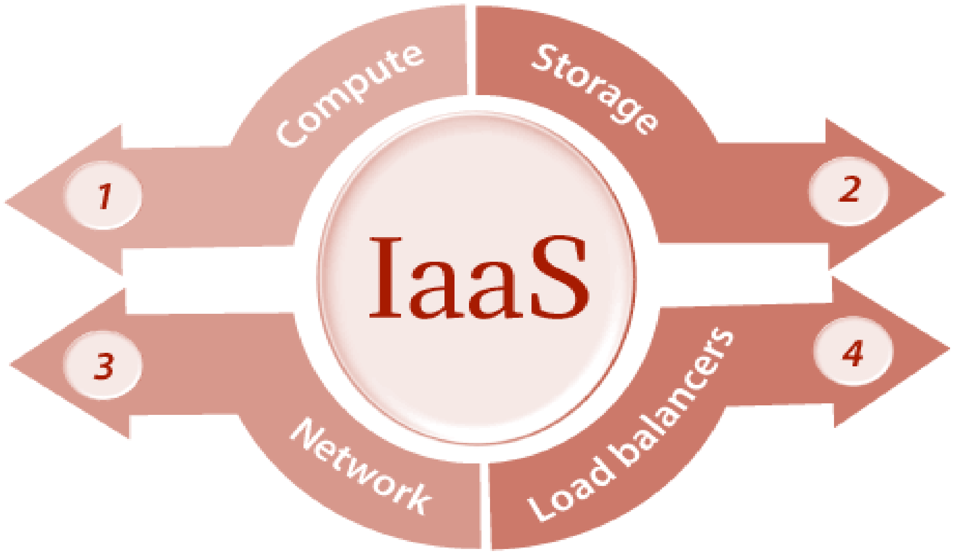
- ❖ SaaS model provides the software applications as a service. As a result, **on the customer side, there is no upfront investment in servers or software licensing.**
- ❖ **Customer data is stored in the cloud** that is either vendor proprietary or a publically hosted cloud supporting the PaaS and IaaS.
- ❖ On the provider side, **costs are rather low, compared with conventional hosting of user applications.**
- ❖ Microsoft online sharepoint and CRM software from Salesforce.com are good examples.



Cloud Service Models

❑ Infrastructure-as-a-Service (IaaS)

- ❖ This model allows users to rent processing, storage, networks, and other resources.
- ❖ User can deploy and run the guest OS and applications.
- ❖ User does not manage or control the underlying cloud infrastructure but has control over OS, storage, deployed applications, and possibly select networking components. This IaaS model encompasses the storage as a service, computation resource as a service, and communication resource as a service.

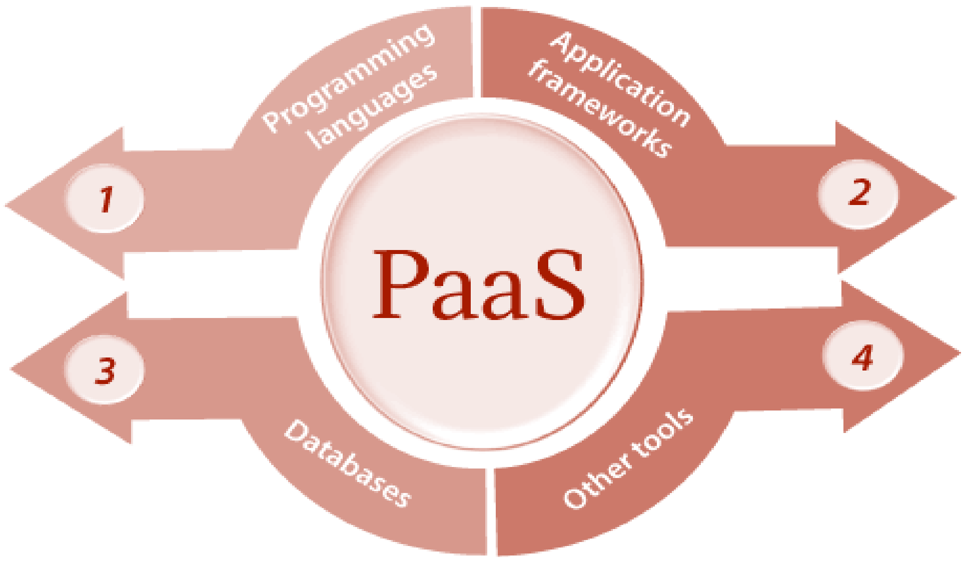


- ❖ Examples of IaaS providers include AWS EC2, Azure Virtual Machines, and Google Compute Engine.

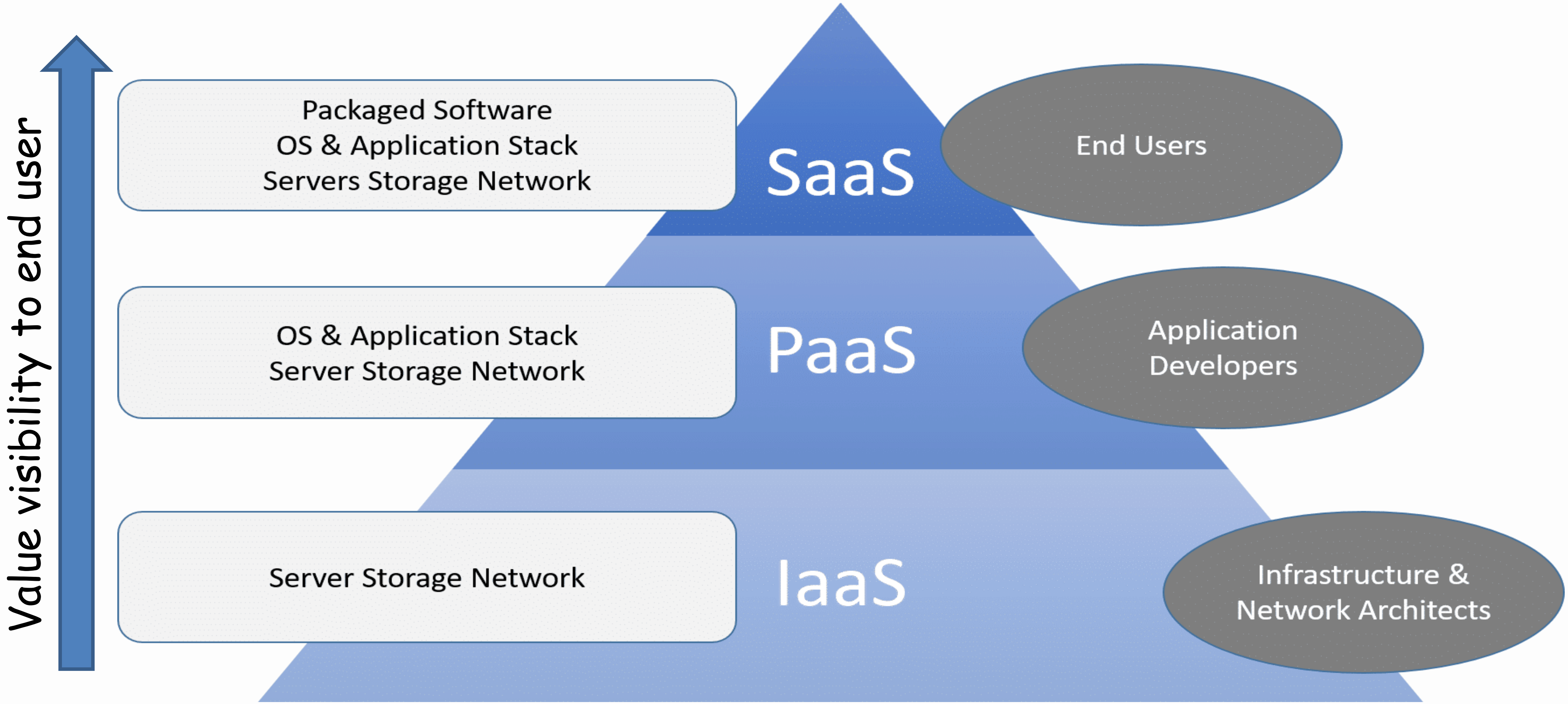
Cloud Service Models

❑ Platform-as-a-Service (PaaS)

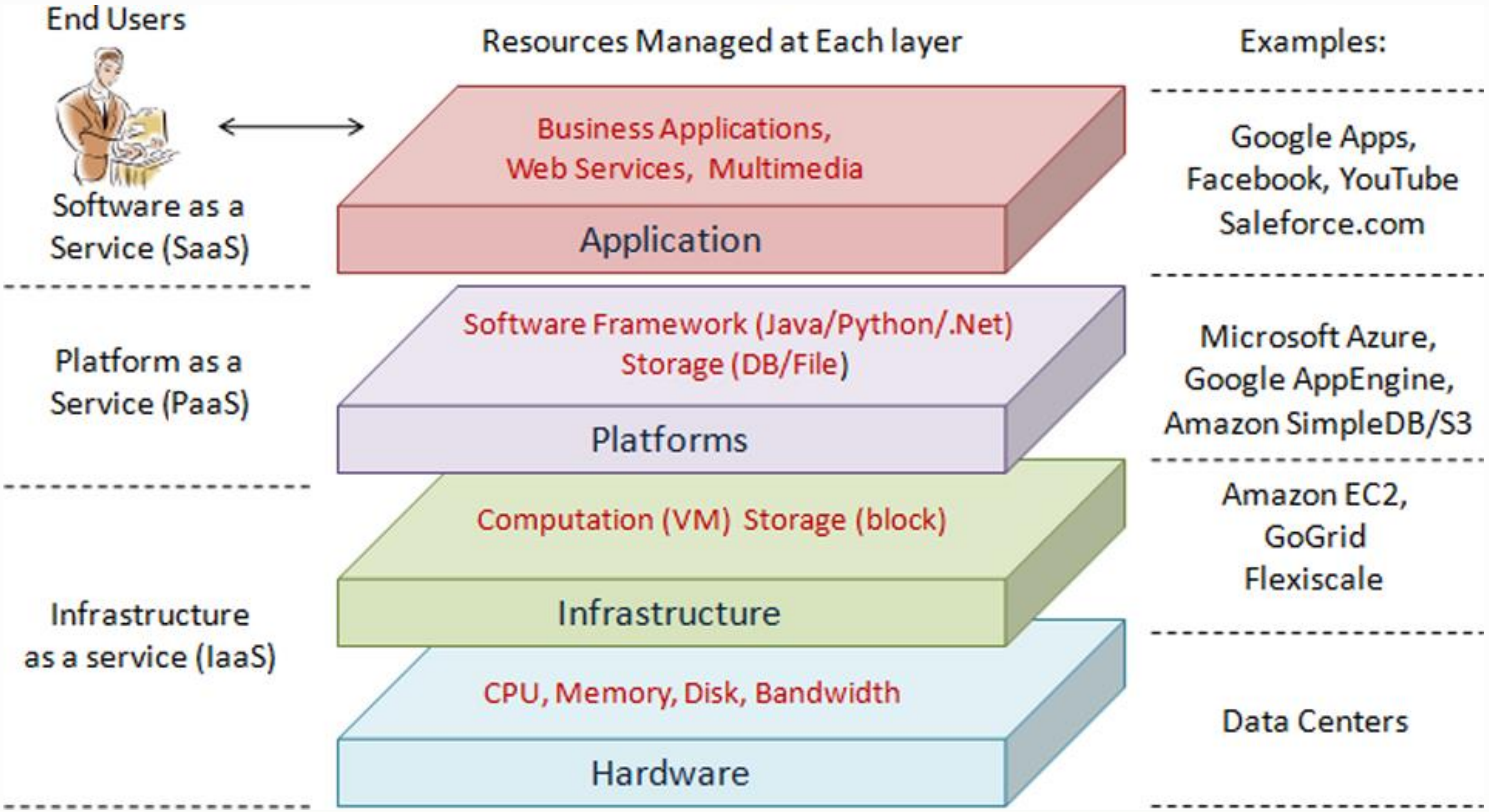
- The PaaS model provides the user to deploy user-built applications on top of the cloud infrastructure, that are built using the programming languages and software tools supported by the provider (e.g., Java, python, .Net).
- User does not manage the underlying cloud infrastructure.
- **Cloud provider facilitates to support the entire application development, testing and operation support** on a well-defined service platform.
- E.g. Google App Engine.



Cloud Service Models

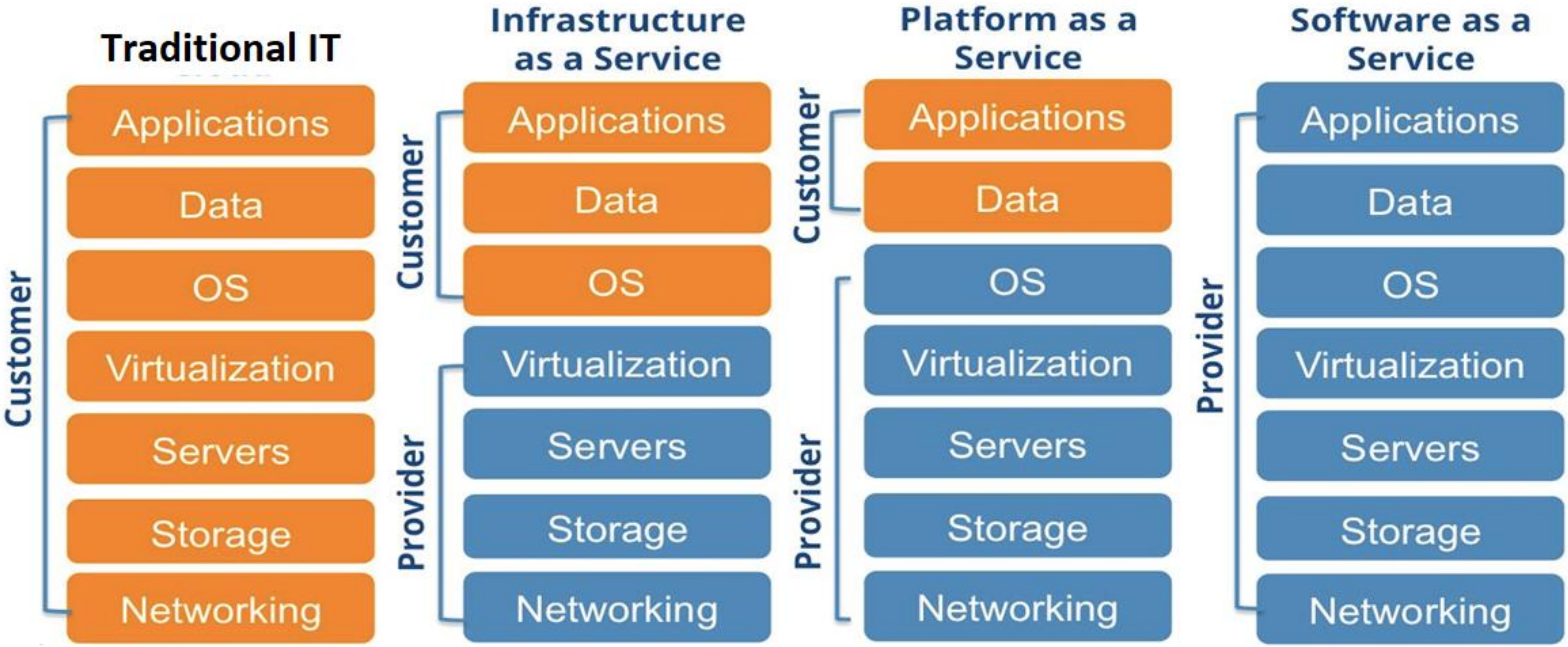


Cloud Layer Infrastructure



Cloud Service Models

- User levels of control over assets and services:

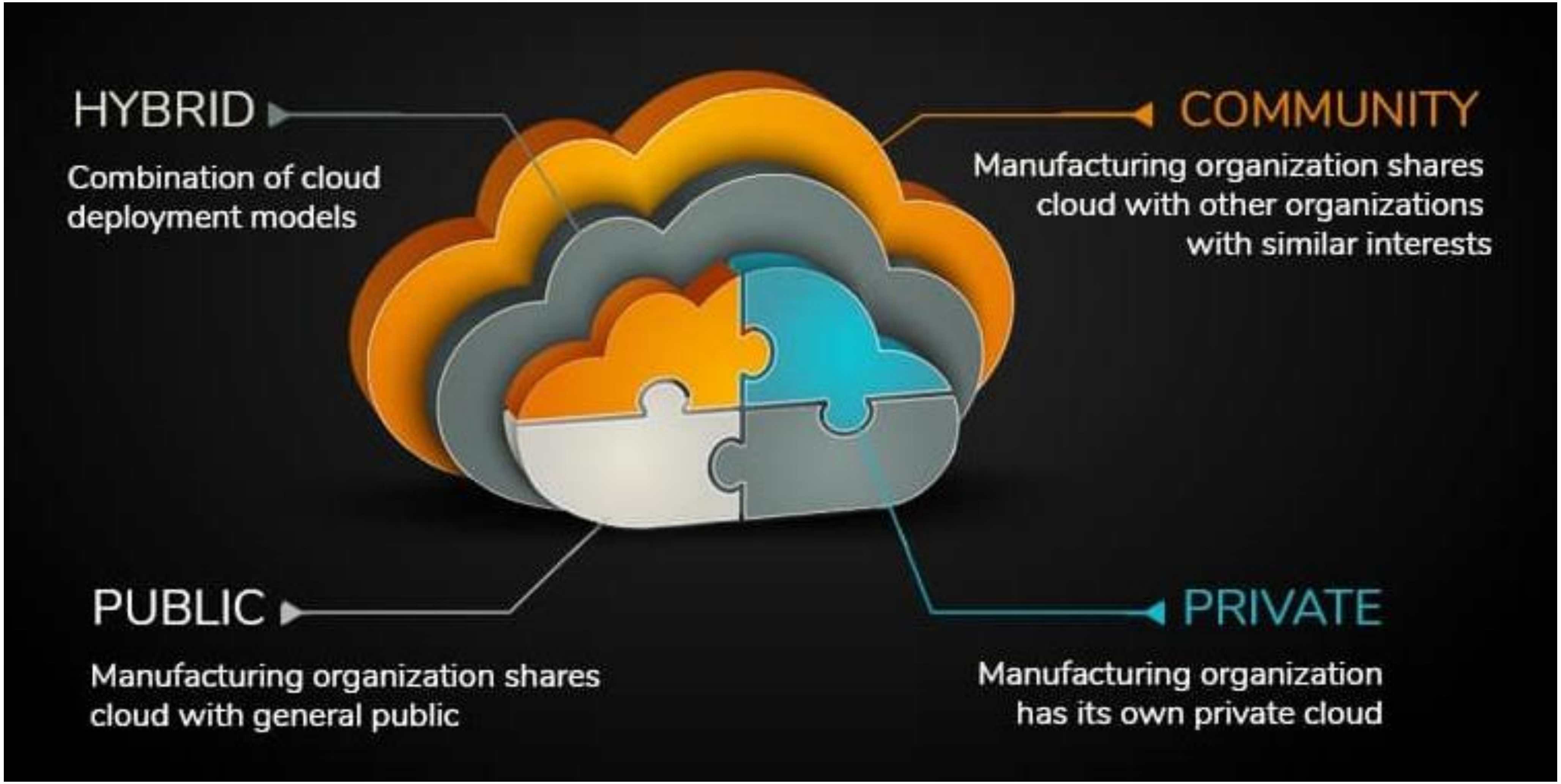


Cloud Service Models

- Everything-as-a-Service (**XaaS**)
  - Composite second level services
    - Security-as-a-Service
      - McAfee:
        - -- McAfee SaaS Email Archiving
        - -- McAfee SaaS Email Inbound Filtering
        - -- McAfee Vulnerability Assessment SaaS (PEN Tests)
    - CaaS – Communication-as-a-Service
      - VoIP, private PBX

Cloud Deployment Models

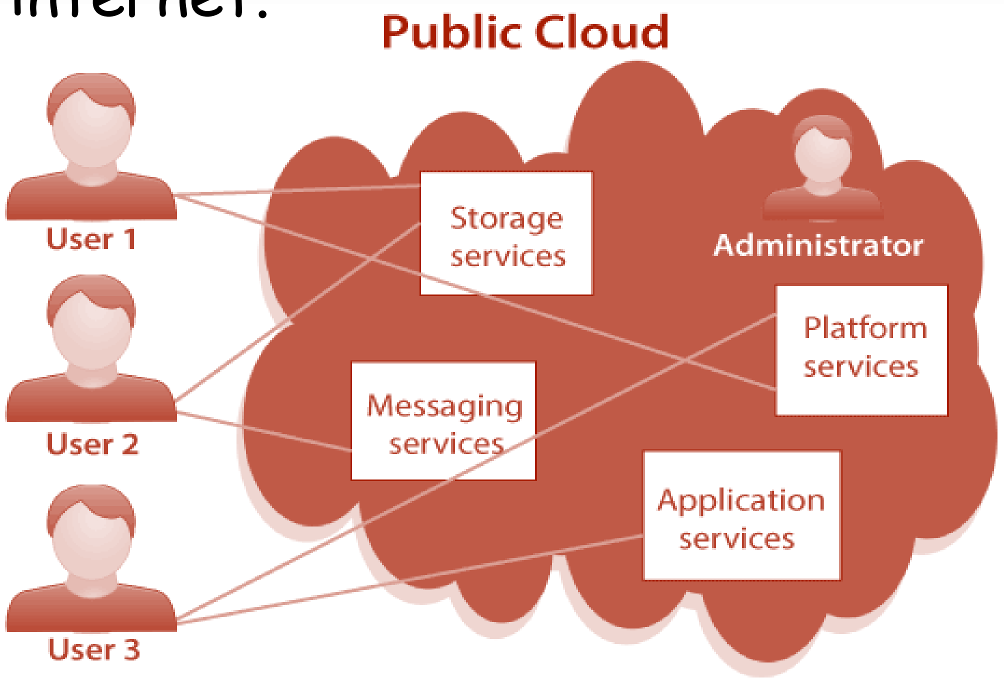
Multiple clouds coexist:



Cloud Deployment Models

❑ Public Cloud:

❖ A public cloud is a type of cloud computing in which a third-party service provider makes computing resources—including anything from ready-to-use software applications, to individual virtual machines (VMs), to complete enterprise-grade infrastructures and development platforms—available to users over the public internet.



Cloud Deployment Models

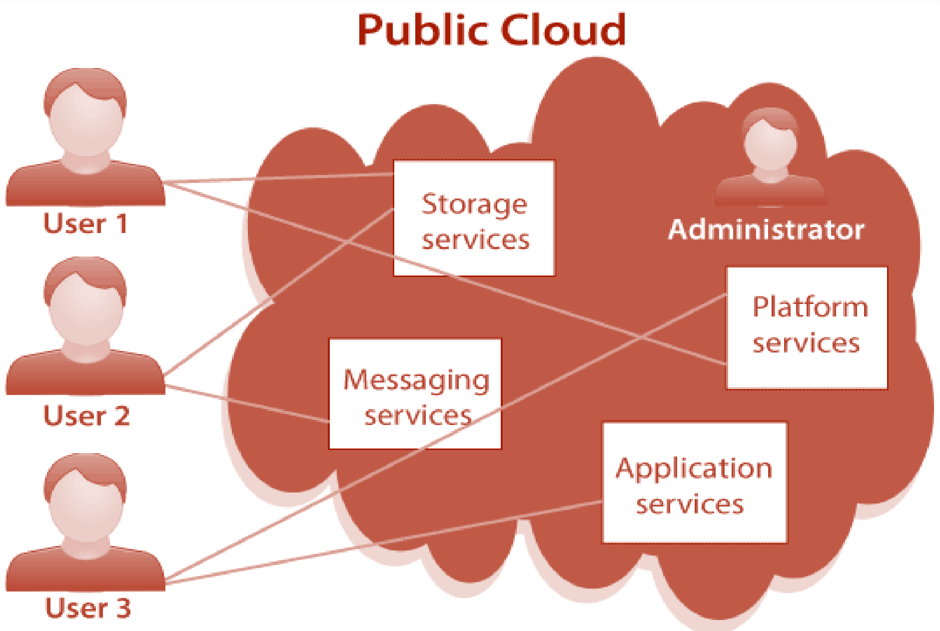
❑ Public Cloud:

- Is a type of cloud computing in which a service provider makes computing resources—including anything from ready-to-use software applications, to individual virtual machines, to complete infrastructures and development platforms—available to users over the public internet.

- A public cloud is a fully virtualized environment that relies on high-bandwidth network connectivity to transmit data.

- Providers have a multi-tenant architecture that enables users -- or tenants -- to run workloads on shared infrastructure and use the same computing resources.

- A tenant's data in the public cloud is logically separated and remains isolated from the data of other tenants.



Cloud Deployment Models

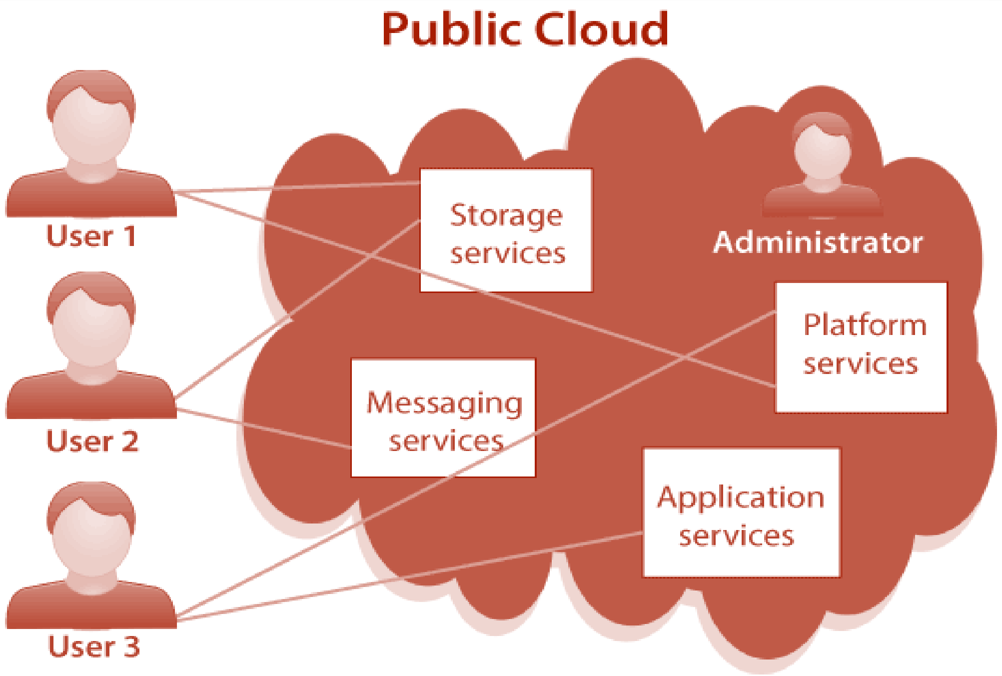
❑ Public Cloud:

- ❖ Providers operate cloud services in logically isolated locations within public cloud regions. These locations, called availability zones consist of two or more connected, highly available physical data centers.

- ❖ Public cloud services may be free or offered through a variety of subscription or on-demand pricing schemes, including a pay-per-usage model

- ❖ It is a cost-effective option for businesses and individuals looking for scalability and flexibility.

- ❖ Public cloud providers, such as AWS, Azure, and GCP, offer a wide range of services accessible to the general public.

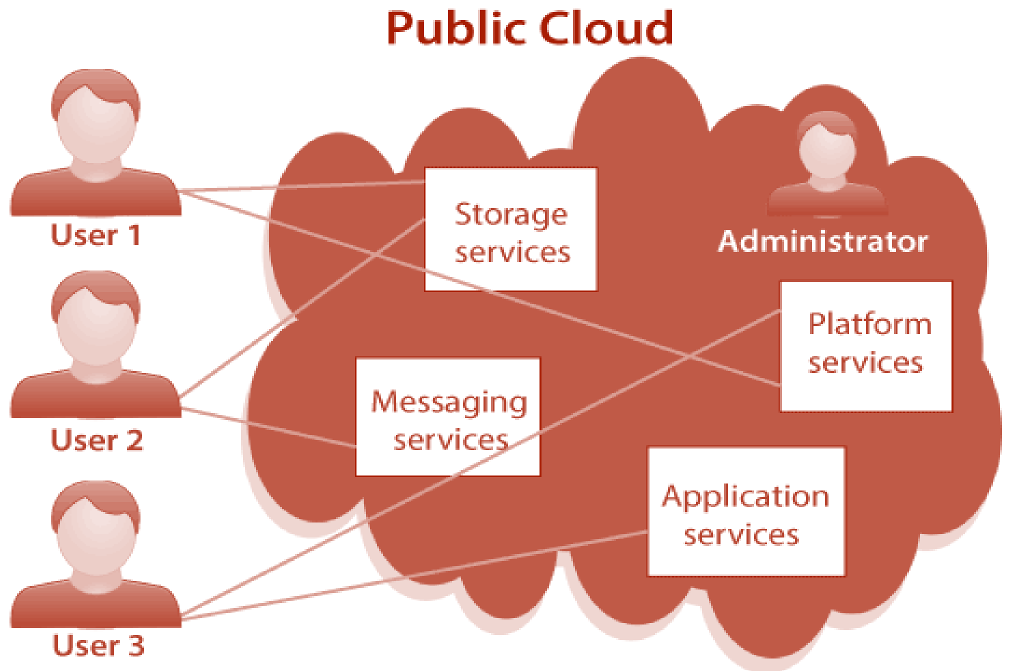


Cloud Deployment Models

❑ Public Cloud:

❖ The main benefits of the public cloud:

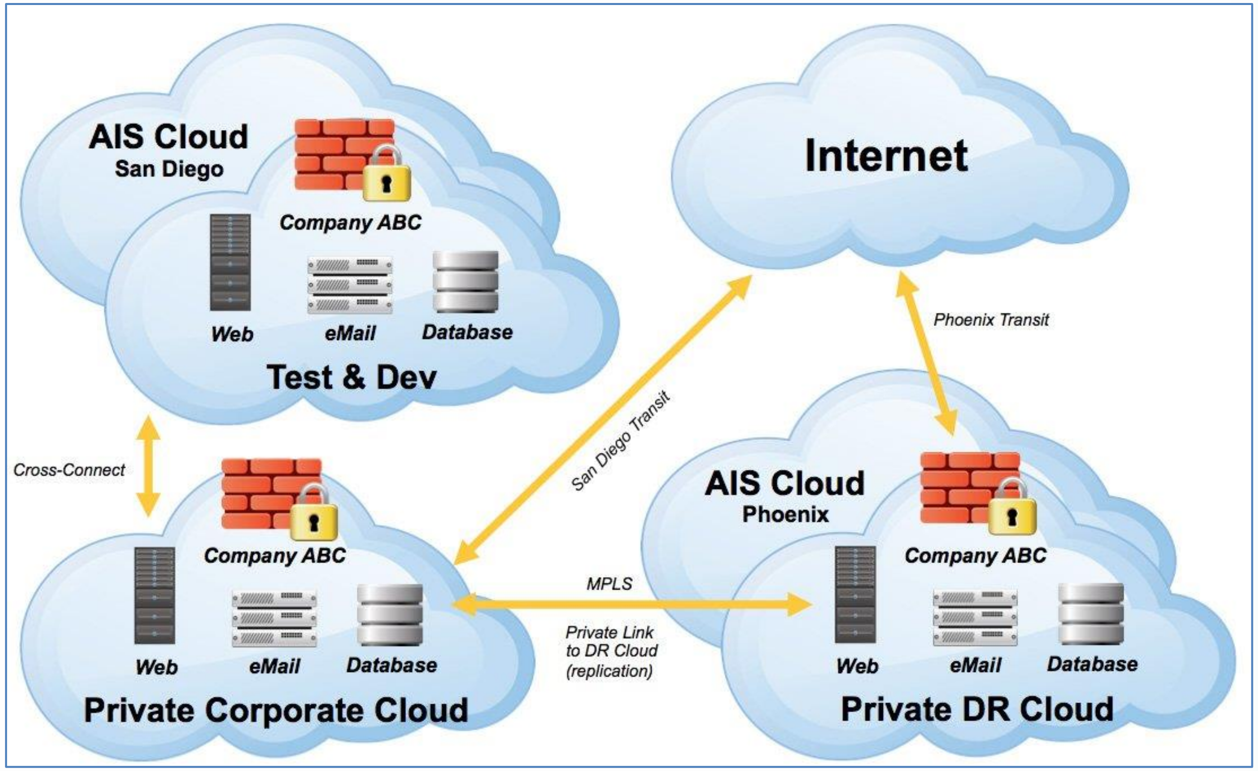
- ➢ Reduced need for organizations to invest in and maintain their own on-premises IT resources;
- ➢ Scalability to meet workload and user demands; and
- ➢ Fewer wasted resources because customers only pay for what they use.



Cloud Deployment Models

❑ Private Cloud:

- Private cloud is cloud infrastructure operated exclusively to a single organization.
- Private cloud environments are designed to meet specific security, compliance, or performance requirements.
- Typically, a private cloud is hosted on-premises, behind the client company's own firewall, but it can also be hosted on a dedicated cloud provider or third-party infrastructure. In either event, the client company has exclusive, isolated access to the infrastructure.
- They offer enhanced control, customization, and privacy but require significant upfront investment.



AIS .. Microsoft Azure Infrastructure solutions
DR ..Disaster Recovery

Cloud Deployment Models

❑ Private Cloud:

Why a private cloud might be chosen:

- ❑ **Enhanced Security & Privacy:** A private cloud is dedicated to a single organization, meaning all data, applications, and resources are isolated from others. This isolation significantly reduces the risk of security breaches compared to a public cloud, where resources are shared among multiple users. It's particularly beneficial for organizations that handle sensitive data or must comply with strict privacy regulations (e.g., healthcare, finance).

- ❑ **Greater Control:** Organizations have full control over their private cloud environment. This allows them to customize infrastructure, configurations, and management policies according to their specific needs. For example, businesses can implement their own security protocols, choose hardware configurations, and set up their own monitoring and reporting tools.

- ❑ **Compliance & Regulatory Requirements:** Many industries are subject to strict regulatory requirements for data storage, processing, and transmission. A private cloud allows organizations to meet these standards by providing the ability to implement customized security measures, data handling procedures, and audit trails, making it easier to ensure compliance.

Cloud Deployment Models

❑ Private Cloud:

Why a private cloud might be chosen:

- ❑ **Customization & Flexibility**: Unlike public clouds, which are designed for general use, a private cloud can be specifically tailored to the needs of the organization. This includes choosing the right applications, integrating with existing IT systems, and configuring the cloud environment to fit operational requirements, performance needs, and workflow demands.

- ❑ **Performance & Reliability**: With a private cloud, an organization doesn't have to share resources with others, which can lead to more predictable and reliable performance. Private clouds can be optimized for specific workloads, ensuring faster response times and lower latency for critical business applications.

- ❑ **Data Control & Sovereignty**: Private clouds enable organizations to retain full control over their data and infrastructure. This is particularly important for businesses that need to ensure that their data stays within specific geographical regions or national boundaries due to data sovereignty laws. For example, certain countries or industries may have regulations requiring that data remain within national borders.

Cloud Deployment Models

❑ Private Cloud:

Why a private cloud might be chosen:

- ❑ **Cost Control & Predictability:** While private clouds can have higher upfront costs than public clouds (due to the need for dedicated hardware and infrastructure), they can provide more predictable operational costs over time. Once the infrastructure is set up, ongoing maintenance and operation costs can be easier to manage, especially for organizations with large-scale or specialized needs.

- ❑ **Scalability** : A private cloud, when built using a flexible architecture, allows organizations to scale resources as needed. Though it may not offer the same elasticity as a public cloud, it can still scale within the boundaries of its infrastructure, often with higher performance and more control over how resources are allocated.

Cloud Deployment Models

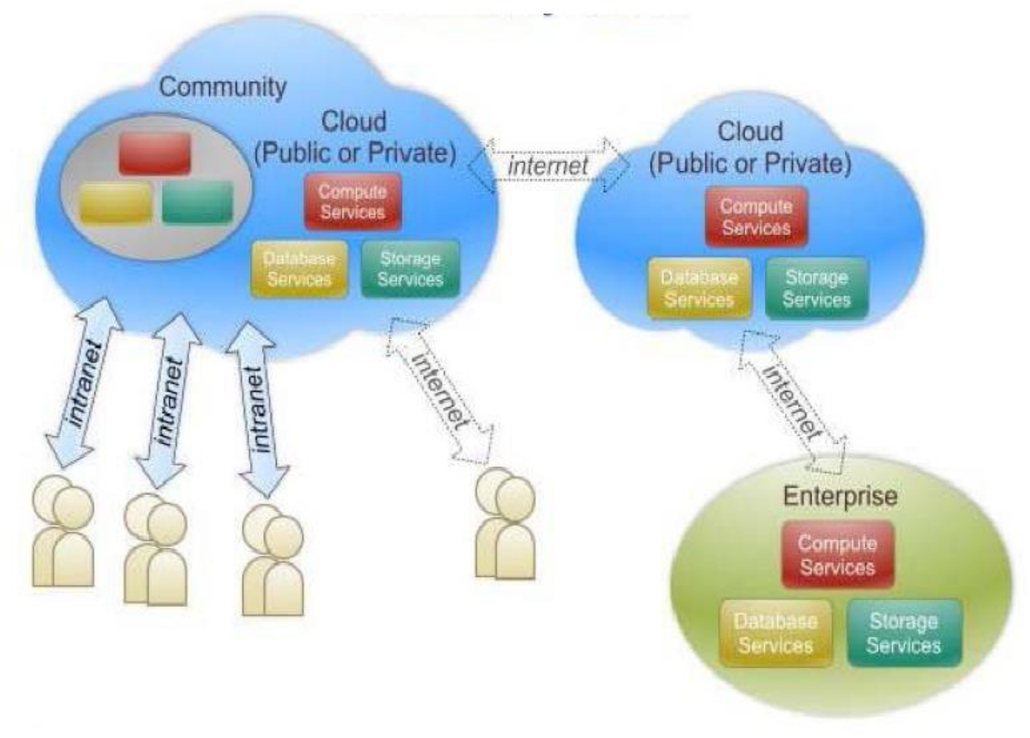
❑ Community Cloud:

- A model where a shared infrastructure is provisioned for exclusive use by a specific community of consumers from organizations with joint concerns, such as mission, security requirements, policy, and compliance considerations.

- It is a hybrid form, combining the privacy and control of a private cloud with the collaborative environment of a public cloud.

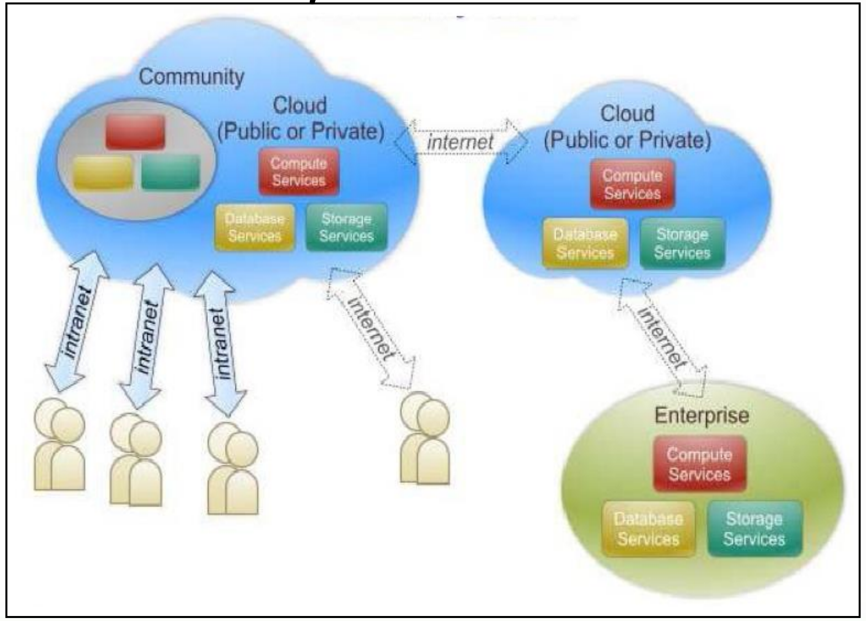
- Community clouds are multi-tenant platforms enabling different organizations to work on a shared platform without compromising security or compliance.

- The primary purpose is to allow multiple customers to work on joint projects and applications that belong to the community.



Cloud Deployment Models

- Community cloud is a form of similar to a private cloud regarding exclusivity.
- Instead of being exclusive to only one organization, it is used by multiple organizations.
- The motivations for sharing a private cloud infrastructure could stem from the shared needs of the organizations in the community.
- It could be that the companies operate in a field with similar needs for security and redundancy in their IT infrastructure; or that the companies share a need for certain kind of compliance.
- The cloud should fulfill the following criteria in order to be called community cloud:
    - Openness
    - Community
    - Graceful failure
    - Convenience and control
    - Environmental sustainability
- The managing of the community cloud could be complex, due to technological complexity presented by the differences in IT infrastructure of the community members.



Cloud Deployment Models

- **Openness:** refers to being free from vendor and technology lock-in.
- **Community:** refers to the communal ownership and shared responsibility.
- **Graceful:** failure is the independence of any single organization; the effects of possible outages in one organization are limited only to that entity and thus don't hinder the availability of the cloud to others.
- **Convenience and Controls:** refer to the distributed control over the resources; no single community member has excess power over the others.
- **Environmental Sustainability:** is made possible by the concentration of resources to the community cloud, which lessens the need for the community members to run their own infrastructure

Cloud Deployment Models

❑ Community Cloud:

Drawbacks :

- **Complex governance:** Managing a shared environment requires complex governance structures, often leading to delays and conflicts.

- **Limited customization:** Due to collective agreements, individual organizations may have less flexibility to tailor the environment to their needs.

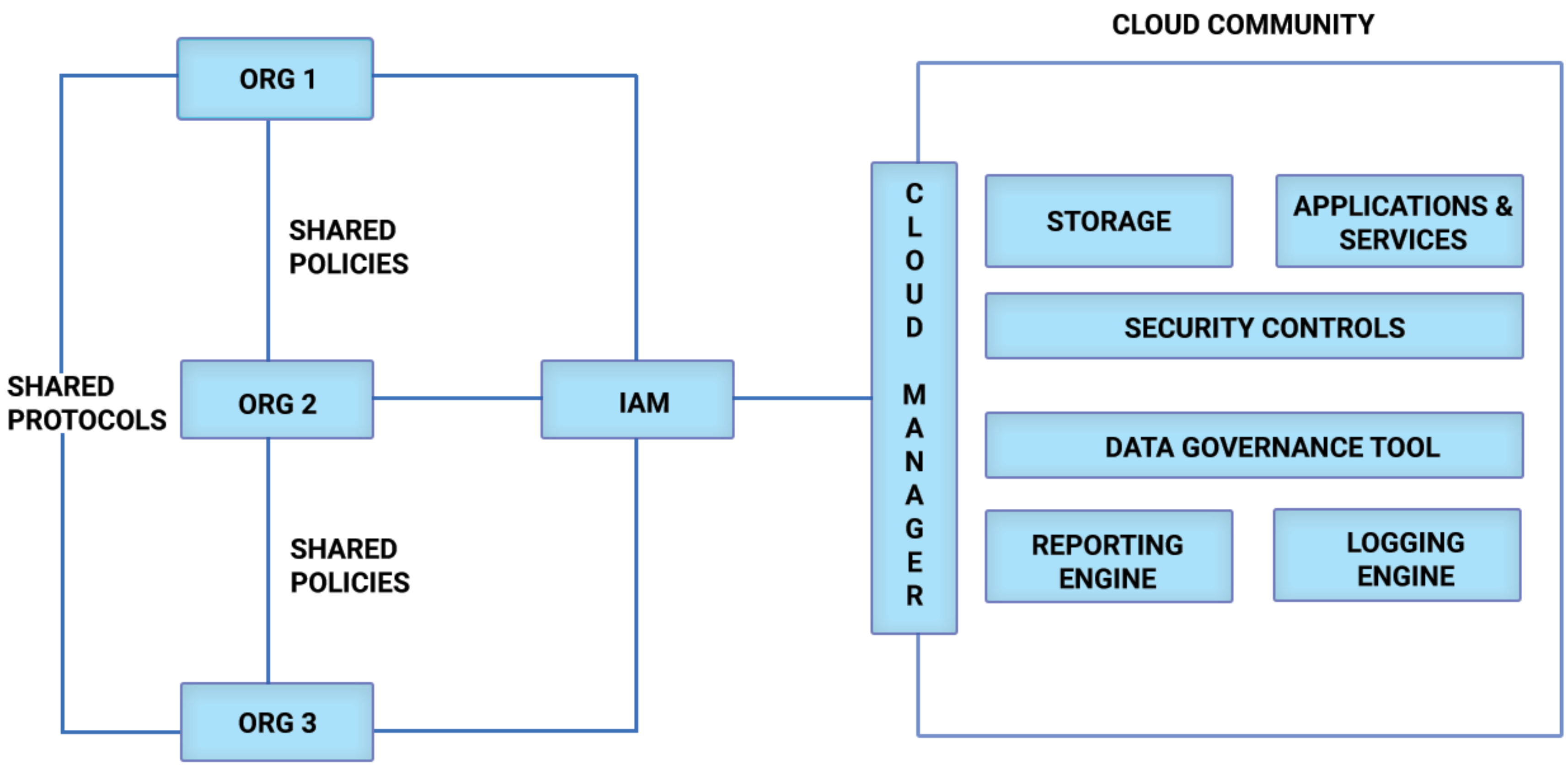
- **Dependency on partners:** The performance and reliability of the cloud depend on all participating organizations adhering to agreed-upon standards and practices.

- **Exit barriers:** Exiting the community cloud can be challenging, involving complex data migration and potential legal implications.

Cloud Deployment Models

❑ Community Cloud:

Components of Community Cloud Architecture:

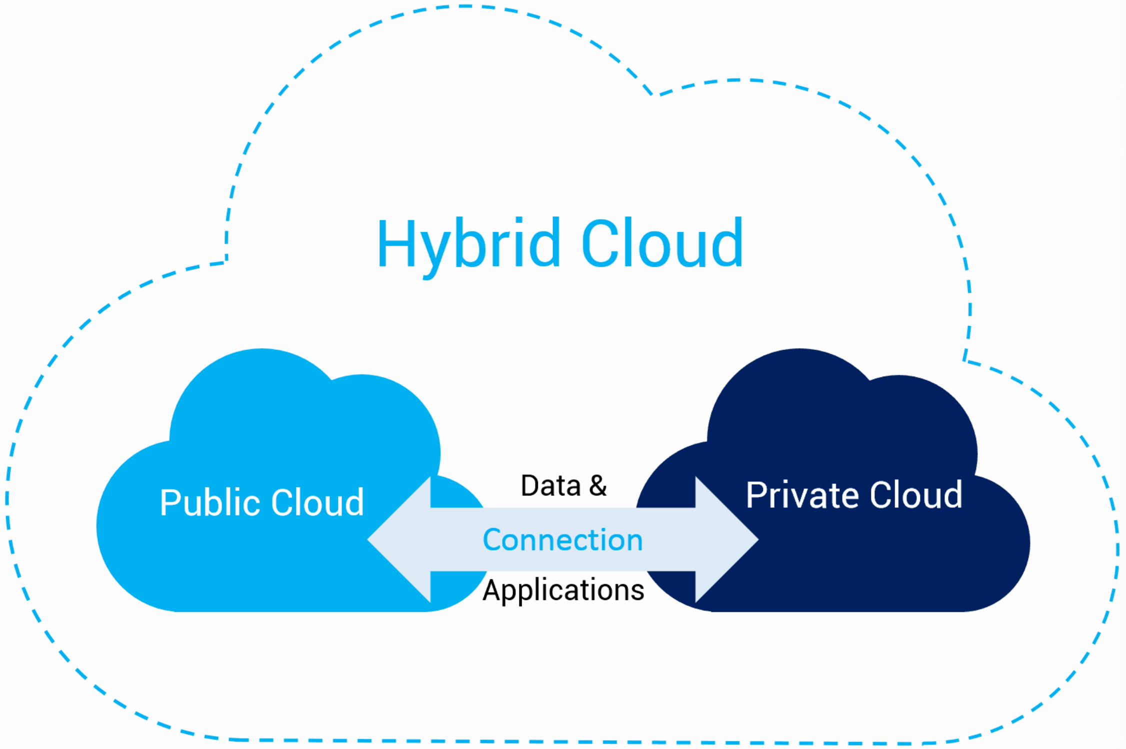


IAM … Identity Access Management

Cloud Deployment Models

❑ Hybrid Cloud:

- ❖ Combines public and private cloud environments, allowing for flexibility and data sharing between the two.
- ❖ Organizations can leverage the benefits of both public and private clouds, ensuring optimal resource allocation.
- ❖ Hybrid cloud deployments enable workload portability and seamless integration between different environments.



Cloud Deployment Models

❑ Final Conclusions:

- **Community clouds:** are more expensive than public clouds but also more secure. Each member of the cloud is allocated a fixed amount of data storage and bandwidth, making scalability somewhat more difficult than with private and public clouds.

- **Public clouds:** are perfect for fledgling companies.

- **private clouds:** are a good fit for large enterprises.

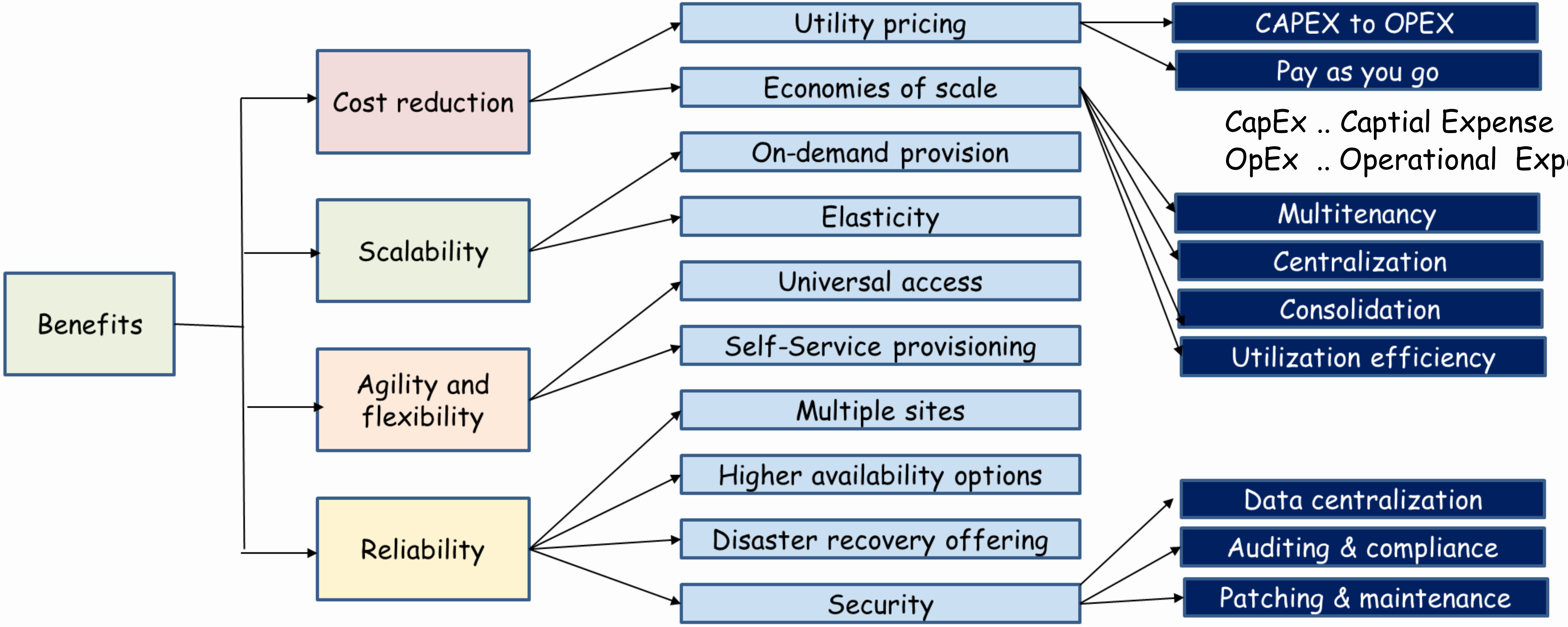
- **Community clouds:** are a great solution for growing organizations in the health, financial, legal, and educational sectors. This is because these industries are the most bound by various regulations.

- **Community cloud implementation:** is more complicated than other types of clouds. This is because of the number of players involved. Decisions are no longer standalone

- **Hybrid cloud:** infrastructure allows you to retain your on-premise, physical data centers, and servers while giving remote employees the flexibility to gain from cloud-hosted resources. However, security can be a major concern for organizations when adopting the hybrid cloud. Distributed assets, diverse hosting environments, and disparate network architectures can make it difficult to achieve a single pane of truth and centralize security governance

Benefits of Cloud Computing

Business perspective: Note, benefits will vary depending on several factors including use case, workload, cloud provider, capabilities, and so on.



Benefits of Cloud Computing

Business perspective: Note, benefits will vary depending on several factors including use case, workload, cloud provider, capabilities, and so on.

	CapEx	OpEx
Time Period	Long-term investments	Short-term, recurring expenses
Purpose	For gaining future benefits	For maintaining daily operations
Examples of Payables	Land, Buildings, Machines, Large-scale IT systems, etc.	Rent, Labor, Utilities, Office Supplies, etc.
On the income statement	CapEx items show up only partially as depreciation	All the OpEx expenses appear in your income statement

Benefits of Cloud Computing

Development and Test

Traditional

High deployment costs to deliver software

- Reduced installation and administration costs
- Lower TCO by improved utilization of software assets

Control and governance chaos in software processes

- Better governance through standardized delivery of services
- Preconfigured software embodying best practices

Onramp and on-boarding of teams reduces time to software delivery

- Tools can be provisioned in minutes. No download, installation or setup.
- Self-administered portal to access to software resources for a globally distributed team

Benefits of Cloud Computing

IT Benefits from Cloud Computing

		Traditional	
Increasing speed and flexibility	Test provisioning	Weeks	Minutes
	Change management	Months	Days/hours
	Release management	Weeks	Minutes
	Service access	Administered	Self-service
	Standardization	Complex	Reuse/share
	Metering/billing	Fixed cost	Variable cost
Reducing costs	Server/storage utilization	10-20%	70-90%
	Payback period	Years	Months

Categories of Cloud Computing Risks

Less Control

Many companies and governments are uncomfortable with the idea of their information located on systems they do not control. Providers must offer a high degree of security transparency to help put customers at ease.

Technology Immaturity

Lack of world-wide adopted Standards.
Use of closed proprietary technologies.
Lack of knowledge and trust.
API Jungle.
Legal uncertainties.

Data Security

Migrating workloads to a shared network and compute infrastructure increases the potential for unauthorized exposure. Authentication and access technologies become increasingly important.

Vendor Lock-in

Interoperability constraints.
Low level of portability of application and services based on cloud.
Contract and exit strategies
Limitations on sharing or transferring data

Compliance

Complying with regulations may prohibit the use of clouds for some applications. Comprehensive auditing capabilities are essential.

Security Management

Providers must supply easy controls to manage firewall and security settings for applications and runtime environments in the cloud.

Reliability

High availability will be a key concern. IT departments will worry about a loss of service should outages occur. Mission critical applications may not run in the cloud without strong availability guarantees.

Cloud Concepts and Technologies

- Virtualization
- Load Balancing
- Scalability and Elasticity
- Deployment
- Replication
- Monitoring
- Identity and Access Management (IAM)
- Service Level Agreements (SLAs)
- Billing